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void vbell();
void dis仁_moves();
char main_screen();
int maxx, maxy, sqxlen=30, sqylen=30;
int fleft, ftop, sqx, sqy, jmoves=0;
float midx, midy;
void far *image[16];
void main(void)
{
    int gdriver=DETECT, gmode, error=
code;

    /* auto-detect the graphics driver
and mode */
    initgraph(&gdriver, &gmode, "");
    errorcode = graphresult(); /* check
for any errors */
    if (errorcode != grOk)
    {
        printf("Graphics error: %s\n",
grapherrormsg(errorcode));
        printf("Press any key to halt:");
        getch();
        exit(1);
    }
    maxx = getmaxx();
    maxy = getmaxy();
    midx = maxx/2;
    midy = maxy/2;
    fleft = midx-2*sqxlen;
    ftop = midy-2*sqylen;

    game_description();
    clearviewport();
    show_screen();
    show_flag();
    while((mkey = main_screen())
!= 'E')
    {
        switch(mkey)
        {
            case 'I' :
                gameinst();
                settextstyle(0, 0,
01);
                setcolor(04);
                outtextxy(100,130,"
<Press any key continue >");
                getch();
                clearviewport();
                break;
            case 'F' :
                show_screen();
                show_flag();
                break;
            case 'D' :
                demo_play();
                break;
            case 'S' :
                simple_play();
                break;
            case 'H' :
                hard_play();
                break;
            case 'E' :
                closegraph();
                exit(1);
                break;

            default: vbell(); break;
        }
    }
    char main_screen()
    {
        char vkey;
        settextstyle(0, 0, 02);
        setcolor(61);
        outtextxy(160,20,"NUMBER PUZZLE
GAME");
        settextstyle(0, 0, 01);

```

```

        setcolor(02);
        rectangle(100,10,580,220);
        outtextxy(200,50,"Main menu");
        settextstyle(0, 0, 01);
        setcolor(6);
        outtextxy(100,70," I. To see Instruc-
tions of the game ");
        outtextxy(100,90," F. To show front
side of game play screen and ");
        outtextxy(100,100," Indian flag
screen");
        outtextxy(100,120," D. A demo of the
game");
        outtextxy(100,140," S. To start play-
ing the simple game");
        outtextxy(100,160," H. To start play-
ing the hard game");
        outtextxy(100,180," E. To Exit the
game");
        setcolor(04);
        outtextxy(100,200,"<Select option
by pressing keys I or F or D or S or H
or E>");
        vkey = getch();
        vkey = toupper(vkey);
        clearviewport();
        return vkey;
    }

    void game_description()
    {
        settextstyle(0, 0, 02);
        setcolor(61);
        outtextxy(200,20,"NUMBER PUZZLE
GAME");
        settextstyle(0, 0, 01);
        setcolor(02);
        rectangle(100,50,580,230);
        outtextxy(100,60," Description:");
        settextstyle(0, 0, 01);
        setcolor(6);
        outtextxy(100,70," This game is just
to test your imagination and logic.");
        outtextxy(100,80," At first you will
be given a 16 squares number game");
        outtextxy(100,90," with numbers in
random fashion. You have to arrange");
        outtextxy(100,100," them in se-
quential fashion by the movement of
green");
        outtextxy(100,110," colored square.
if you finish arrangement with in ");
        outtextxy(100,120," 50 moves you wins
the game.");
        // outtextxy(100,130," .....");
        outtextxy(100,140," Initially you
will be shown the actual squares that
");
        outtextxy(100,150," appear on the
front side and the indian flag that
is");
        outtextxy(100,160," on the back side.
If you arrange numbers successfully");
        outtextxy(100,170," you get the indi-
an flag displayed correctly else you");
        outtextxy(100,180," loose the game
and get the sprinkled indian flag. ");
        outtextxy(100,190," Use main menu to
play with. Good luck.");
        settextstyle(0, 0, 01);
        setcolor(04);
        outtextxy(140,210," <Press any key to
continue>");
        getch();
        clearviewport();
    }

    void gameinst()
    {
        settextstyle(0, 0, 02);
        setcolor(61);
        outtextxy(200,20,"NUMBER PUZZLE

```

```

GAME");
        settextstyle(0, 0, 01);
        setcolor(02);
        rectangle(100,50,580,140);
        outtextxy(100,60," Instruction:");
        settextstyle(0, 0, 01);
        setcolor(6);
        outtextxy(100,70," 1. Use arrow keys
to move green colored astrisk box ");
        outtextxy(100,80," 2. You cannot move
this box outside the border big box");
        outtextxy(100,90," 3. Arrange all the
numbers in sequential order");
        outtextxy(100,100," Like 1 2 3 4 5
by moving green box");
        outtextxy(100,110," 4. Use Esc key
to exit");
    }
    void show_screen()
    {
        clearviewport();
        settextstyle(0, 0, 02);
        setcolor(61);
        outtextxy(200,110,"NUMBER PUZZLE
GAME");
        settextstyle(0, 0, 01);
        outtextxy(280,140," SAMPLE");
        outtextxy(260,160,"PLAY GAME
SCREEN");
        settextstyle(0, 0, 01);
        setcolor(65);
        rectangle(fleft-5, ftop-
5, fleft+4*(5+sqxlen),
ftop+4*(5+sqylen));
        rectangle(fleft-7, ftop-
7, fleft+4*(7+sqxlen)-5, ftop+4*(7+sqylen)-
5);
        init_boxes(s);
        draw_boxes(s);
        settextstyle(0, 0, 01);
        setcolor(04);
        outtextxy(230,350," <Press any key to
continue>");
        getch();
        clearviewport();
    }

    void show_flag()
    {
        settextstyle(0, 0, 02);
        setcolor(61);
        outtextxy(200,110,"NUMBER PUZZLE
GAME");
        settextstyle(0, 0, 01);
        outtextxy(280,140," SAMPLE");
        outtextxy(280,160,"FLAG SCREEN");
        setcolor(65);
        rectangle(fleft-5, ftop-
5, fleft+4*(5+sqxlen),
ftop+4*(5+sqylen));
        rectangle(fleft-7, ftop-
7, fleft+4*(7+sqxlen)-5, ftop+4*(7+sqylen)-
5);
        settextstyle(0, 0, 01);
        setfillstyle(01,63);
        flag_india(s);
        setcolor(04);
        outtextxy(230,350," <Press any key to
continue>");
        getch();
        settextstyle(0, 0, 01);
        clearviewport();
    }

    void simple_play()
    {
        gameinst();
        init_boxes(s);
        shuffle_boxes(s,esnum);
        settextstyle(0, 0, 01);
        setcolor(04);
        outtextxy(280,150," GIVEN");

```